

Operation and Maintenance Manual

Diesel engine-generator sets

DP00275D5SFA

DP00300D5SFA

DS00300D5SFA

DS00330D5SFA

Application groups 3B, 3D

Built in Europe

Control versions 2, 3a, 3b, 4

SS180011/02E



System designation	Engine model	Application group
DP00275D5SFA	6R1600G10F	3B, Continuous operation, variable
DP00300D5SFA	6R1600G20F	3B, Continuous operation, variable
DS00300D5SFA	6R1600G70F	3D, Short-term operation, fuel stop
DS00330D5SFA	6R1600G80F	3D, Short-term operation, fuel stop

Table 1: Applicability

Preface

This Operation & Maintenance Manual provides general instructions for operating your MTU Onsite Energy engine-generator set properly. It is essential that every person who works on or with the engine-generator set be completely familiar with the contents of this manual and that he/she carefully follows the instructions contained herein.

Each operation and maintenance may require some modification of the suggested guidelines in this manual. They must be consistent with locally applicable standards and take into consideration safety guidelines and measures.

Following this guide will result in an efficient and reliable installation.

Carefully follow all procedures and safety precautions to ensure proper equipment operation and to avoid bodily injury. Read and follow the Safety section at the beginning of this manual.

All instructions and diagrams have been checked for accuracy and simplicity of application, however, the skills of the operator are most important. MTU Onsite Energy does not guarantee the result of any operation contained in this manual nor can MTU Onsite Energy assume responsibility for any injury or damage to property. Persons engaging in operating do so entirely at their own risk.

In this publication the manufacturer of the engine-generator set is known as MTU Onsite Energy.

Table of Contents

1 Safety		2.6.9 Sub-base fuel tank	72
1.1 Important requirements for all products	8	2.7 Air System	74
1.2 Personnel and organizational requirements	9	2.7.1 Air system	74
1.3 Safety requirements for startup and operation	10	2.7.2 Air filter	76
1.4 Safety requirements for maintenance and repair work	11	2.7.3 Heavy duty air filter - Mounting location	78
1.5 Fire prevention and environmental protection, fluids and lubricants, auxiliary materials	14	2.7.4 Heavy duty air filter	79
1.6 Standards for safety notices in the text	16	2.8 Exhaust System	81
2 Functional Description		2.8.1 Exhaust system	81
2.1 Chapter overview	17	2.8.2 Exhaust silencer	82
2.2 Engine-Generator Set	18	2.9 Cooling System	83
2.2.1 Engine-generator set - General information	18	2.9.1 Cooling system	83
2.2.2 Engine-generator set - Scope of supply	20	2.9.2 Coolant cooler	84
2.3 Engine	23	2.9.3 Preheater - Mounting location	86
2.3.1 MTU 6 R 1600 Gx0 engine	23	2.9.4 Preheater	87
2.3.2 Sensors and actuators	26	2.10 Electrical System	89
2.3.3 Engine side and cylinder designations	28	2.10.1 Starter battery	89
2.3.4 Firing order	29	2.10.2 Battery charger	91
2.4 Generator	30	2.10.3 Circuit breaker	93
2.4.1 Generator - General information	30	2.11 Base and mounting system	96
2.4.2 Analog voltage regulator	34	2.12 Enclosure	97
2.5 Monitoring and Control	36	3 Technical Data	
2.5.1 Control version 2 - Isolated operation (without mains)	36	3.1 Technical data DP00275D5S (fuel optimized)	99
2.5.2 Control version 3a - With automatic transfer switch	38	3.2 Technical data DP00275D5S (exhaust optimized TA-Luft)	101
2.5.3 Control version 3b - With electrical interlock circuit breakers	40	3.3 Technical data DP00300D5S (fuel optimized)	103
2.5.4 Control version 4 - Parallel operation (island)	42	3.4 Technical data DP00300D5S (exhaust optimized TA-Luft)	105
2.5.5 Control cabinet with DGC-2020 - General information	44	3.5 Technical data DS00300D5S (fuel optimized)	107
2.5.6 Control panel DGC-2020 - Display operations	50	3.6 Technical data DS00330D5S (fuel optimized)	109
2.5.7 Smart Connect	54	4 Transport	
2.5.8 Read out the active control version on DGC-2020	56	4.1 Transport	111
2.6 Fuel System	57	5 Operation	
2.6.1 Fuel system	57	5.1 Chapter overview	113
2.6.2 Fuel prefilter	59	5.2 Lockout/tagout procedure	114
2.6.3 Fuel prefilter with water separator - Mounting location	61	5.3 Unlocking procedure	117
2.6.4 Fuel prefilter with water separator (single)	63	5.4 Putting the engine-generator set into operation	119
2.6.5 Fuel prefilter with water separator (double)	65	5.5 Start procedures - Overview	122
2.6.6 Fuel cooler - Mounting location	67	5.6 Stop procedures - Overview	123
2.6.7 Fuel cooler	68		
2.6.8 Day tank	70		

5.7 Start and Stop Procedures for Control Version 2	124	8.3.3 Cylinder head cover – Removal and installation	263
5.7.1 Automatic start and stop	124	8.3.4 Fuel system – Venting	264
5.7.2 Manual start	126	8.3.5 Fuel filter – Replacement	265
5.7.3 Manual stop	127	8.3.6 Engine oil level – Check	266
5.8 Start and Stop Procedures for Control Version 3a	128	8.3.7 Engine oil – Change	267
5.8.1 Automatic start and stop with ATS	128	8.3.8 Engine oil filter – Replacement	268
5.8.2 Manual start	130	8.3.9 Engine coolant – Level check	269
5.8.3 Manual stop	131	8.3.10 Engine coolant – Change	270
5.9 Start and Stop Procedures for Control Version 3b	132	8.3.11 Engine coolant – Draining	271
5.9.1 Automatic start and stop with electrical interlock circuit breakers	132	8.3.12 Engine coolant – Filling	273
5.9.2 Manual start	135	8.3.13 Engine coolant pump – Relief bore check	275
5.9.3 Manual stop	136	8.3.14 Drive belt – Adjustment	276
5.10 Start and Stop Procedures for Control Version 4	137	8.3.15 Drive belt – Condition check	277
5.10.1 Automatic start and stop	137	8.3.16 Drive belt – Tension check	278
5.10.2 Manual start	139	8.3.17 Drive belt – Replacement	279
5.10.3 Manual stop	141	8.3.18 Engine wiring – Check	280
5.11 Emergency stop	142	8.3.19 Engine governor and connectors – Cleaning	281
5.12 Exercising checks	143	8.3.20 Engine governor – Checking plug connections	282
5.13 Putting the engine-generator set out of operation	145	8.4 Generator	283
6 Troubleshooting		8.4.1 Generator – Check	283
6.1 Chapter overview	146	8.4.2 Generator – Wiring check	284
6.2 Troubleshooting by symptoms	147	8.4.3 Generator mounting securing screws – Security check	285
6.3 Engine-Generator Set Fault Codes	152	8.5 Engine-Generator Set Control Panel	286
6.3.1 Recording of fault codes on DGC-2020	152	8.5.1 Control panel DGC-2020 – Backup battery replacement	286
6.3.2 List of fault codes on DGC-2020 in alphabetical order	154	8.6 Fuel System	287
6.4 Engine Fault Codes	212	8.6.1 Fuel prefilter with water separator (double) – Differential pressure check	287
6.4.1 Fault codes on Smart Connect display	212	8.6.2 Fuel prefilter – Draining overview	288
6.4.2 List of MTU fault codes in numerical order (ADEC-ECU8)	213	8.6.3 Fuel prefilter – Draining	289
7 Maintenance		8.6.4 Fuel prefilter with water separator – Draining	290
7.1 Maintenance task reference table [QL1]	257	8.6.5 Fuel prefilter – Replacement overview	291
8 Task Description		8.6.6 Fuel prefilter – Replacement	292
8.1 Chapter overview	258	8.6.7 Fuel prefilter with water separator – Filter element replacement	293
8.2 Engine-Generator Set	259	8.7 Air System	295
8.2.1 Engine-generator set – Test run	259	8.7.1 Service indicator – Signal ring position check	295
8.3 Engine	260	8.7.2 Air filter – Replacement	296
8.3.1 Engine – Barring with starting system	260	8.7.3 Air filter – Removal and installation	297
8.3.2 Valve clearance – Check and adjustment	261	8.7.4 Heavy duty air filter – Removal and installation	298
		8.8 Cooling System	299
		8.8.1 Coolant cooler – Check for leaks and contamination	299
		8.8.2 Coolant cooler – Core face cleaning	300
		8.8.3 Preheater – Coolant line check	302
		8.8.4 Preheater – Heating element housing wiring check	303
		8.9 Electrical System	304
		8.9.1 Circuit breaker – Check and function test	304

9 Appendix A		9.4 MTU contact persons/service partners	329
9.1 Abbreviations	306	10 Appendix B	
9.2 Conversion tables	309	10.1 Special Tools	330
9.3 Glossary	313	10.2 Consumables	333
		10.3 Index	334

1 Safety

1.1 Important requirements for all products

Nameplate

The product is identified by a nameplate, model designation or serial number which must match the information on the title page of this manual.

Nameplate, model designation or serial number can be found on the product.

General information

This product may pose a risk of injury or damage in the following cases:

- Improper use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications
- Noncompliance with the safety instructions and warning notices

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/MTU contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (including engine management/parameters)
- In compliance with all safety regulations and in accordance with all warning notices in this manual
- With maintenance work performed in accordance with the (→ Maintenance Schedule) throughout the useful life of the product
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair
- By contracting only workshops authorized by the manufacturer to carry out repair and overhaul

Any other use shall be considered non-intended. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

Changes or modifications

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

TM-ID: 0000040530 - 004

1.2 Personnel and organizational requirements

Organizational measures of the operator

This manual must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair and transport personnel at all times.

Use this manual as a basis for instructing personnel on product operation and repair, whereby the safety-relevant instructions, in particular, must be read and understood.

This is particularly important in the case of personnel who only occasionally perform work on or around the product. This personnel must be instructed repeatedly.

Personnel requirements

All work on the product shall be carried out by trained and qualified personnel only.

- Training at the Training Center of the manufacturer
- Qualified personnel specialized in mechanical and plant engineering

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair and transport.

Working clothes and personal protective equipment

Wear proper protective clothing for all work.

When working, always wear the necessary personal protective equipment (e.g. ear protectors, protective gloves, goggles, breathing protection). Observe the information on personal protective equipment in the respective activity description.

1.3 Safety requirements for startup and operation

Safety requirements for startup

The product must be installed and accepted in accordance with manufacturers' specifications prior to initial startup.

All the requisite regulatory approvals must have been granted and all startup requirements fulfilled prior to initial startup.

Whenever the product is started, always ensure that:

- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating machine components.
- Wearers of cardiac pacemakers or other active medical implants are well clear.
- The operating room is adequately ventilated.
- Exhaust pipework is leak-tight and routed to atmosphere.
- Battery terminals, generator terminals and cables are guarded to preclude accidental contact.
- All personnel is clear of the danger zone represented by moving parts.

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems are working properly.

Safety requirements for operation

The operator must be familiar with the controls and displays.

The operator must be aware of the consequences of any operations he/she performs.

During operation, the display instruments and monitoring units must be constantly observed in regard of present operating status, limit value violation and warning or alarm messages.

Malfunctions and emergency stop

Emergency procedures, in particular emergency stop, must be practiced on a regular basis.

Take the following steps if a system malfunction is detected or signaled by the system:

- Inform the duty supervisor(s).
- Evaluate the message.
- Respond to the emergency appropriately, e.g. execute an emergency stop.

Operation

Do not stay in the operating room when the product is running unless absolutely necessary and then as briefly as possible.

Keep a safe distance away from the product whenever possible. Do not touch the product unless expressly instructed to do so.

Do not inhale exhaust gases emitted by the product.

Ensure that the following requirements have been fulfilled before starting the product:

- Ear protectors are worn.
- Mop up any leaked or spilled fluids and lubricants immediately or soak up with a suitable binding agent.

Operation of electrical equipment

Some components are live (high voltage) when electrical equipment is in operation.

Observe the safety instructions for these appliances.